

Schlachtberger et al.: Formulating an Electricity Planning LP

From the planning problem to the value of transmission

1. Start with the planning problem

Reading: Schlachtberger, D. P., Brown, T., Schramm, S., and Greiner, M. (2017). “The benefits of co-operation in a highly renewable European electricity network.” *Energy* 134: 469–481. [Public manuscript](#).

European countries must supply electricity in every hour while sharply reducing carbon emissions. Wind and solar are low-emission sources, but their available output varies with weather and time of day. Three ways to cover a local shortfall are to generate dispatchable electricity, discharge stored energy, or import electricity from another country. Each requires infrastructure and incurs costs or uses a limited resource.

The planning question: What generation, storage, and international transmission capacities should be built, and how should they operate, to meet demand at minimum annual cost subject to an emissions limit?

The paper represents 30 countries with one node per country, hourly demand and weather from 2011, and a CO₂ cap 95% below 1990 electricity-sector emissions. The network connects countries along the existing topology. The model chooses capacities and operations together. It does not prescribe a wind or solar share and then calculate its cost.

Separate choices from information

Demand, renewable availability, costs, efficiencies, connection lengths, resource potentials, and policy caps are **given**. Infrastructure capacities, hourly production, storage operation, and electricity flows are **chosen**.

Existing hydro and pumped-hydro capacities are fixed in the paper. New hydro capacity is not an investment choice.

Define variables before writing equations

Use n for countries, s for **generation and storage technologies**, t for hours, and ℓ for international connections, as in the paper.

Variable	Meaning	Units
$G_{n,s}$	Generation or storage power capacity	MW
$g_{n,s,t}$	Generation or net storage dispatch	MW
$soc_{n,s,t}$	Storage state of charge	MWh
$F_{\ell}, f_{\ell,t}$	Line capacity and signed flow	MW

Storage uses the same $G_{n,s}$ and $g_{n,s,t}$ notation as generation. For storage, positive dispatch supplies the grid; negative dispatch charges it.

2. Translate electricity supply into constraints

“Demand must be met in every country in every hour”

Let $d_{n,t}$ be demand. Define $K_{n\ell} = +1$ where positive flow leaves country n , -1 where it enters, and zero otherwise. Then

$$\underbrace{\sum_s g_{n,s,t}}_{\text{generation and net storage dispatch}} - d_{n,t} = \underbrace{\sum_\ell K_{n\ell} f_{\ell,t}}_{\text{net exports}} \quad \forall n, t.$$

This is the paper’s equation (2). Charging enters $g_{n,s,t}$ negatively; discharging enters positively. Exports reduce supply available for domestic demand. Conservation requires equality.

Check the signs. If country A generates 80 MW, serves 60 MW of demand, and charges storage at 5 MW, it can export 15 MW. At country B, that same flow enters with the opposite sign. Summing balances over countries cancels international flows: transmission redistributes electricity rather than creating it.

Why not impose just an annual balance? Annual production can equal annual demand while production falls short on a particular evening. Every hour must be feasible, including hours of high demand and little wind.

“A generator cannot produce more than its available capacity”

Let $\bar{g}_{n,s,t}$ denote available output per MW of installed capacity:

$$0 \leq g_{n,s,t} \leq \bar{g}_{n,s,t} G_{n,s} \quad \forall n, s, t.$$

These bounds apply to generators, not signed storage dispatch. For gas turbines, use $\bar{g} = 1$ in this formulation. For wind and solar, availability is determined by weather.

Connection to the R example: North’s wind availability is 0.4 in period 1 and 1.0 in period 2. These fixed inputs encode less favorable wind in the first period and conditions permitting full output in the second. The same 5 MW installation can therefore produce **up to 2 MW**, then **up to 5 MW**. The model chooses capacity and actual output; it does not choose the weather.

Output may be below the available amount. The inequality permits **curtailment** when additional renewable output cannot be used or stored. Requiring equality would force all available renewable energy onto the grid.

“Suitable locations are limited”

$$0 \leq G_{n,s} \leq G_{n,s}^{\max}.$$

The upper bound represents geographic installation potential. Omit a finite upper bound where the model assumes no such limit; fix capacity for existing hydro facilities. These restrictions apply to installed capacity, while weather availability restricts its hourly use.

Formulation check: $g \leq \bar{g}G$ is linear. The weather coefficient $\bar{g}$ is known. Rewriting it as $g - \bar{g}G \leq 0$ makes the linear structure explicit.

3. Translate storage, networks, and policy limits

“Electricity can be shifted across hours, with losses”

The paper denotes charging and discharging efficiencies by η_1 and η_2 . Its equation (6) is

$$\begin{aligned} soc_{n,s,t} = soc_{n,s,t-1} &+ \eta_1 g_{n,s,t,\text{charge}} - \eta_2^{-1} g_{n,s,t,\text{discharge}} \\ &+ g_{n,s,t,\text{inflow}} - g_{n,s,t,\text{spillage}}. \end{aligned}$$

Time steps are one hour: each power term implicitly multiplies one hour to give energy. Charging at 10 MW with $\eta_1 = 0.9$ adds 9 MWh; delivering 9 MWh with $\eta_2 = 0.9$ removes 10 MWh from storage.

For a storage technology, the dispatch in the electricity balance is $g_{n,s,t} = g_{n,s,t,\text{discharge}} - g_{n,s,t,\text{charge}}$. Both component rates are nonnegative and bounded by $G_{n,s}$.

$$0 \leq soc_{n,s,t} \leq E_{n,s} = h_{s,\max} G_{n,s}, \quad soc_{n,s,0} = soc_{n,s,T}.$$

The fixed duration $h_{s,\max}$ is 6 hours for batteries and 168 hours for hydrogen. A 100 MW battery therefore has 600 MWh of energy capacity. The cyclic condition prevents a free annual withdrawal from the initial stock.

Natural inflow and spillage apply to reservoir hydro; electrical charging is zero there. Reservoir and pumped-hydro capacities are fixed. Batteries and hydrogen have no natural inflow. No binary charge/discharge switch is used.

“Each connection has a carrying capacity”

$$-F_\ell \leq f_{\ell,t} \leq F_\ell \quad \forall \ell, t, \quad F_\ell \geq 0.$$

Flows can reverse direction. Capacity cannot be negative. Two linear inequalities implement the bound $|f_{\ell,t}| \leq F_\ell$. The paper uses a controllable transport network, without modeling the full electrical power-flow physics or bottlenecks within each country.

“Total international transmission development is restricted”

$$\sum_{\ell} l_{\ell} F_{\ell} \leq \text{CAP}_{LV}.$$

Line length l_{ℓ} is fixed. The cap CAP_{LV} is measured in MWkm; a 1,000 MW connection spanning 500 km uses 500,000 MWkm. This is a physical development limit, separate from the construction costs in the objective.

“Total carbon emissions cannot exceed a budget”

$$\sum_{n,s,t} \frac{e_s}{\eta_s} g_{n,s,t} \leq \text{CAP}_{\text{CO}_2}.$$

Here e_s is tonnes of CO_2 per MWh of fuel and η_s is generation efficiency. Electrical output divided by efficiency gives fuel use. For non-emitting generation, $e_s = 0$. The carbon budget limits annual fuel use; it does not directly specify gas capacity or a renewable capacity share.

4. Assemble the LP

“Choose the least-cost feasible electricity system”

The paper uses $c_{n,s}$ for annualized generation and storage capacity costs, c_ℓ for annualized line costs, and $o_{n,s}$ for variable dispatch costs. Storage capacity costs include energy capacity through the fixed ratio $h_{s,\max}$. Existing fixed infrastructure adds constant costs.

Using the paper’s notation, the objective and constraints are

$$\begin{aligned}
& \min_{G,F,g,f,soc} \quad \sum_{n,s} c_{n,s} G_{n,s} + \sum_{\ell} c_{\ell} F_{\ell} + \sum_{n,s,t} o_{n,s} g_{n,s,t} \\
& \text{subject to} \quad \sum_s g_{n,s,t} - d_{n,t} = \sum_{\ell} K_{n\ell} f_{\ell,t}, \\
& \quad 0 \leq g_{n,s,t} \leq \bar{g}_{n,s,t} G_{n,s} \quad (\text{generators}), \\
& \quad 0 \leq G_{n,s} \leq G_{n,s}^{\max}, \\
& \quad soc_{n,s,t} = soc_{n,s,t-1} + \eta_1 g_{n,s,t,\text{charge}} - \eta_2^{-1} g_{n,s,t,\text{discharge}} \\
& \quad \quad + g_{n,s,t,\text{inflow}} - g_{n,s,t,\text{spillage}}, \\
& \quad g_{n,s,t} = g_{n,s,t,\text{discharge}} - g_{n,s,t,\text{charge}} \quad (\text{storage}), \\
& \quad 0 \leq g_{n,s,t,\text{charge}}, g_{n,s,t,\text{discharge}} \leq G_{n,s}, \\
& \quad 0 \leq soc_{n,s,t} \leq h_{s,\max} G_{n,s}, \quad soc_{n,s,0} = soc_{n,s,T}, \\
& \quad -F_{\ell} \leq f_{\ell,t} \leq F_{\ell}, \quad F_{\ell} \geq 0, \\
& \quad \sum_{\ell} l_{\ell} F_{\ell} \leq \text{CAP}_{LV}, \\
& \quad \sum_{n,s,t} (e_s/\eta_s) g_{n,s,t} \leq \text{CAP}_{\text{CO}_2}.
\end{aligned}$$

Apply generator and storage rows only to the corresponding technologies. Storage component flows are included in the choices summarized by g ; inflow is given and spillage is nonnegative. Fix existing hydro capacities, use $\bar{g} = 1$ for gas, and omit capacity upper bounds where unrestricted. All sums over time use the paper’s implicit one-hour steps.

Check the formulation before solving

1. **Units:** Each balance is in MW; each storage balance is in MWh; emissions are in tonnes; every objective term is in euros per year.
2. **Signs and domains:** Charging and exports use local electricity. Generator output, stocks, and capacities are nonnegative; net storage dispatch and line flows are signed.
3. **Linearity:** Costs, availability, efficiencies, durations, and lengths are parameters. There are no products of decision variables.
4. **Completeness:** Capacity constraints link investments to hourly operation. The storage recurrence links hours. The network links countries.

Stacking variables gives $\min c^\top x$ subject to linear equalities and inequalities. If a solver requires nonnegative variables, write $f = f^+ - f^-$, with $f^+, f^- \geq 0$. The time dimension adds rows and columns; it does not make this a nonlinear program.

Ask students: What becomes possible if we accidentally omit $g_{n,s,t} \leq \bar{g}_{n,s,t} G_{n,s}$? The model could produce electricity without paying for the capacity needed to generate it.

5. Motivate the solution: compare complete ways to meet demand

The cheapest generator in isolation need not be part of the cheapest electricity system. The relevant comparison includes the infrastructure needed to deliver electricity at the time and place of demand.

A two-country thought experiment

Suppose each country needs 1 MW in each of two one-hour periods. Existing wind capacity could produce the following outputs:

Available wind output (MW)	Hour 1	Hour 2
Country A	2	0
Country B	0	2
Combined demand	2	2

This is a teaching example, not a scenario from the paper. With a 1 MW connection, A exports 1 MW in hour 1 and B exports 1 MW in hour 2. No storage or backup is needed in these two hours. Without the connection, each country must bridge its own shortfall using storage or backup. With lossy storage, the given wind energy alone is insufficient over a cyclic two-hour horizon: additional generation is needed to cover storage losses.

Change the example: Put both wind surpluses in hour 1. A connection cannot move electricity to hour 2. Storage or dispatchable generation is now necessary even with unrestricted trade. The value of transmission depends on the spatial correlation of generation, not just its annual sum.

What the optimizer trades off

- **More local renewables:** reduce dependence on imports, but may require storage, backup, or curtailment during high-output hours.
- **More storage:** shift supply across hours, paying capacity costs and losing some energy in each cycle.
- **More transmission:** share geographically diverse output and access productive sites, paying construction costs and using the line-volume cap.
- **More gas generation:** cover difficult hours, but use the emissions budget.

The optimizer evaluates these options jointly across all hours. Building capacity that is attractive in one hour can be unattractive over the year; rare shortfalls can nevertheless justify capacity with low utilization.

The paper solves this sparse LP with PyPSA and Gurobi’s barrier algorithm. Students do not need to reproduce the large solve to understand its logic: the primal solution identifies a feasible least-cost plan, and the dual values measure the costs of tightening its constraints.

Predict before seeing the results

If the transmission cap is relaxed while demand, weather, costs, and the carbon budget remain fixed, minimum total cost cannot rise: the previous solution remains feasible. Predict more use of productive wind locations and less reliance on local storage. Individual countries or technologies need not change monotonically even though minimum total cost does.

6. Follow one sensitivity analysis: relax the transmission cap

The experiment changes only CAP_{LV} in $\sum_{\ell} l_{\ell} F_{\ell} \leq CAP_{LV}$. For each value, **re-optimize all investment and operating decisions**. Keep weather, demand, technology costs, resource bounds, and the CO_2 budget fixed. Do not hold the generation portfolio at its initial optimum.

Scenario	Line volume (TWkm)	Average system cost (EUR/MWh)
No international transmission	0.00	84.1
Existing-grid-equivalent volume	31.25	75.7
Compromise grid	125.00	67.5
Economically optimal grid	285.70	64.8

Source: paper, Table 2. The existing-grid-equivalent scenario optimally allocates the historical aggregate volume; individual line capacities need not match the actual grid. TWkm denotes a million MWkm.

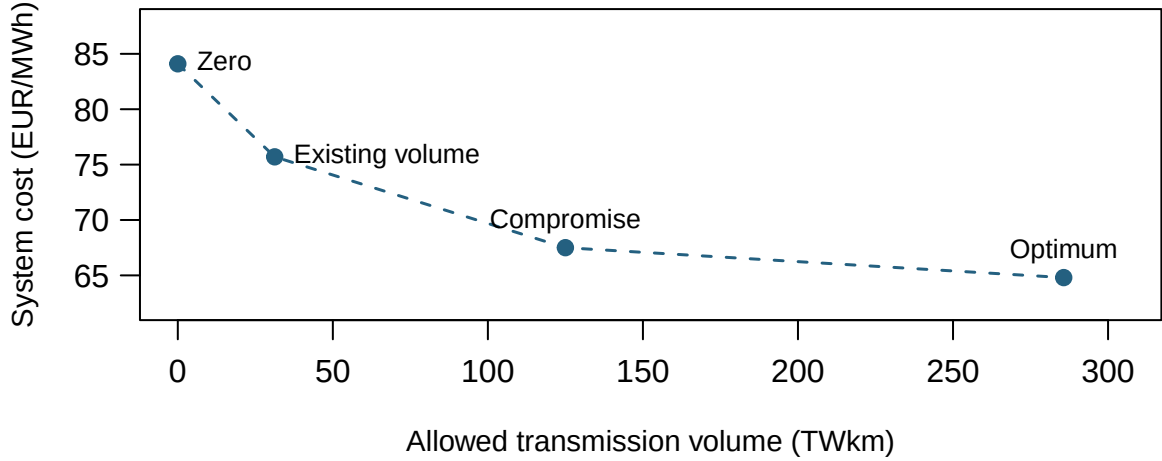


Figure 1: Four published scenarios from Table 2. Dashed segments connect observations as visual guides; they do not reconstruct the full model’s sensitivity curve or its LP breakpoints.

Quantify the compromise

Relative to the zero-transmission benchmark, the maximum reported reduction is $84.1 - 64.8 = 19.3$ EUR/MWh. The compromise captures $84.1 - 67.5 = 16.6$:

$$\text{Savings share} = \frac{16.6}{19.3} \approx 86\%, \quad \text{Volume share} = \frac{125}{285.7} \approx 44\%.$$

The paper reports approximately 85% of savings; 86% follows from its rounded table entries. Against the existing-volume benchmark instead, the savings share is $(75.7 - 67.5)/(75.7 - 64.8) \approx 75\%$. State the benchmark.

Interpretation: Much of the gain is available before the economic optimum. Restricting transmission sharply favors solar and local storage; relaxing the cap permits more wind and geographic balancing. These are endogenous responses to the cap, not imposed changes in technology shares.

7. Use the shadow price to explain the sensitivity result

For the sensitivity derivation, let $C^*(\text{CAP}_{LV})$ be minimized **annual total cost** (teaching notation). Collect the transmission-cap term in the Lagrangian, with nonnegative multiplier μ_{LV} :

$$\mathcal{L} = C(x) + \mu_{LV} \left(\sum_{\ell} l_{\ell} F_{\ell} - \text{CAP}_{LV} \right) + \text{terms for the other constraints.}$$

Where the value function is differentiable, the envelope result gives

$$\frac{dC^*(\text{CAP}_{LV})}{d\text{CAP}_{LV}} = -\mu_{LV}, \quad \Delta C^* \approx -\mu_{LV} \Delta \text{CAP}_{LV}.$$

The multiplier measures annual euros saved by permitting one additional MWkm. Construction costs are already in the objective: this is a **net system saving**, not avoided storage cost alone.

Why the marginal value falls

First, more allowed line volume expands the feasible set, so $C^*(\text{CAP}_{LV})$ is nonincreasing. Second, an LP's value function is piecewise linear and convex in this right-hand-side parameter. Its slope becomes less negative as CAP_{LV} increases; correspondingly, the marginal saving μ_{LV} weakly declines. At a corner there may be multiple optimal dual values, so use the appropriate one-sided slope.

The paper's many-scenario curve can look smooth even though the underlying LP value function is piecewise linear. **A changing slope across scenarios does not make the optimization problem nonlinear.**

Complementary slackness gives

$$\mu_{LV} \left(\text{CAP}_{LV} - \sum_{\ell} l_{\ell} F_{\ell} \right) = 0.$$

Once the cap is above the unconstrained cost-optimal line volume, it is slack and $\mu_{LV} = 0$. Permitting still more capacity has no value because the model can already build its preferred network. A binding constraint does not, by itself, guarantee a strictly positive multiplier.

Finite changes require more than one local shadow price

For a large relaxation, re-solve as the active constraints change:

$$C^*(\text{CAP}_{LV}^{(0)}) - C^*(\text{CAP}_{LV}^{(1)}) = \int_{\text{CAP}_{LV}^{(0)}}^{\text{CAP}_{LV}^{(1)}} \mu_{LV}(u) du.$$

The Table 2 endpoints give finite changes in reported average costs, not raw dual values. Neither a secant slope between those scenarios nor its units should be confused with the annual EUR/MWkm multiplier above. To check a solver's dual numerically, compare annual objective values from nearby re-solves using a consistent MWkm scale and sign convention.

Connection to the previous application: A right-hand-side change alters the feasible set and the optimal allocation. Its value depends on what is allowed to adjust. Here a simple upper-bound relaxation always preserves the old feasible choices; it carries no obligation to build the extra grid.

8. Interpret the result and check understanding

What can be concluded?

Under the paper’s assumptions, international cooperation reduces the cost of meeting a stringent emissions target. It allows countries to share resources and manage variability across space. The benefits diminish, so a grid smaller than the least-cost optimum can retain most of the savings.

The line-volume cap represents a restriction on development. It does not estimate public opposition or environmental damages in euros. Consequently, the compromise is an economically informative alternative, not a demonstrated social optimum after accounting for all external costs.

The magnitude depends on the aggregation of each country to one node, one year of known weather and demand, fixed storage durations, and the technology menu and costs. The model omits internal network bottlenecks, demand response, and flexibility from other energy sectors. These limitations affect the numerical prescription while leaving the LP reasoning useful.

Questions for discussion

1. If we require wind output to equal available output in every hour, which flexibility do we remove? How could that change feasibility?
2. Why can a carbon cap bind even when gas generation capacity is not fully used in most hours?
3. Why must generation and storage be re-optimized when CAP_{LV} changes?
4. If a more permissive transmission cap produces a higher minimized annual cost, what should we check before interpreting the result?
5. Can a country face a higher electricity price after integration even though Europe-wide total cost falls?

Instructor checkpoints

1. Equality removes curtailment. Surplus output must be consumed, stored, or exported; otherwise the constraints can become infeasible.
2. Emissions restrict annual fossil energy, whereas capacity restricts instantaneous power. Gas capacity can serve rare shortage hours.
3. Holding other investments fixed values a different, narrower adjustment. The paper compares complete least-cost systems at each cap.
4. Check that only the upper bound changed, the previous plan remains feasible, both solves reached comparable optima, and the comparison uses the same annual objective rather than differently normalized measures.
5. Yes. An exporting region can face greater demand for its inexpensive resources. Aggregate savings do not establish how gains are distributed.

Source and interpretation notes

- Model: paper Section 2, equations (1)–(10); assumptions: Section 3. Symbols follow the paper, including $G_{n,s}$ for generation and storage capacity, $soc_{n,s,t}$, $\bar{g}_{n,s,t}$, and CAP_{LV} .
- Sensitivity: Section 4.1, Figure 2, and Table 2. The manuscript prose gives 79.9 EUR/MWh for the compromise; these notes use Table 2’s 67.5, which agrees with the stated savings relationship.
- Figure 8 expresses transmission value as an equivalent overnight capital cost. It is not directly the raw annual net multiplier derived here. The notes use the cap experiment and Table 2 to keep that distinction clear.